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# Small Object Detection Using YOLO

## Introduction

YOLO pretrained datasets are great for general object detection, but struggle with small objects. Sometimes detecting these small objects is critical. In this exercise, a pretrained YOLO model is fine tuned to detect traffic lights from a street view camera on a car.

The base pretrained model is yolob8n. The dataset used to fine tune the model is the LISA Traffic Light Dataset [1] complete with YOLO style annotation. Testing the resulting model is done using street view videos from Chico, CA provided by the instructor.

## Training

Fine tuning the pretrained yolov8n model was performed with several variations of hyperparameters to obtain an optimized model. The variations tested include the size of the image passed into the neural network and a suite of additional augmentation not active in the default settings of the YOLO training function.

**Table 1.** Fine-Tuning Model Parameters

| Model Training Parameters | mAP-95 | Precision | Recall |
|---|---|---|---|
| Image size = 640 | 0.896 | 0.972 | 0.979 |
| Image size = 1280 | 0.914 | 0.971 | 0.983 |
| Image size = 1280<br>+ additional augmentation | 0.839 | 0.966 | 0.895 |

Table 1 shows that increasing image size provided better overall results from 640 to 1280. After that step, additional augmentation hyperparameters like rotation, shearing, and perspective shifts were added on top of default data augmentation. These proved to be detrimental, pulling down all major metrics of mAP-95, precision, and recall. Therefore, the model selected for final testing and optimization was the standard training settings with an image size of 1280.

## Testing/Optimizing

Once a good performing fine-tuned model to detect traffic lights was trained, we compared the base pretrained model with that to analyze the performance on small objects. For a rigorous test, a frame with traffic lights in the background was selected from the testing

videos to simulate real usage. A moment in one of the testing videos was selected to differentiate the pretrained model from the variations of the fine-tuned model.

**Fig. 1.** Pretrained YOLOv8n inferencing.

On the pretrained model, object detection is working, but it is clearly not optimized for traffic lights. The traffic lights, very small in the distance, are recognizable for a human, but the model is only identifying one of them. It appears the model only detects one out of about six traffic lights facing the camera.

**Fig. 2.** Fine-Tuned confidence threshold 0.5.

Figure 2 shows the finetuned model operating on the same test video, but now the model is only identifying traffic lights. There is still an issue, however, that not all six of the signals facing the camera are detected; This model is detecting four signals.

**Fig. 3.** Fine-Tuned confidence threshold 0.2.

Pulling the confidence threshold down to 0.2 results in the model detecting the lower two lights that the 0.5 threshold failed to detect. However, there are more sporadic false positives with this configuration. We can see that the lower two signals now detected have confidence of 0.49 and 0.32, which confirms that this was the issue.

## Conclusion

A YOLO model was successfully fine-tuned to detect just traffic lights at smaller scale than the pretrained model was able to. An annotated traffic light dataset from the web was used to do the fine tuning. Tinkering with fine-tuning settings and the confidence threshold resulted in a well performing final model which successfully recognizes

## References

[1] https://www.kaggle.com/datasets/kietzer0/lisa-traffic-light-dataset-yolo?resource=download